

Anurag Pandey

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Education

UPES, Dehradun
B.Tech C.S.E AI/ML

Aug 2023 – May 2027
7.7 CGPA

Professional Summary

Software Engineer and Machine Learning Engineer with hands-on experience building end-to-end applications — from training and deploying ensemble ML models to architecting REST APIs, databases, and browser extensions. Comfortable across Python, Java, and JavaScript with exposure to both full-stack development and production ML pipelines.

Professional Experience

IBM

June 2026 – July 2026

AI/ML Intern

Live: undebuggedbit-chimera-fd.hf.space

- Co-built CHIMERA-FD, a live cascaded fraud detection platform (LightGBM + isotonic recalibration) achieving 92.13% accuracy with zero engine errors on 3,000,000 held-out rows, scale-invariant across three orders of magnitude of test set size.
- Architected end-to-end multi-tenant B2B stack — FastAPI + Next.js 15 + Postgres, JWT + bcrypt auth, single-image Docker on Hugging Face Spaces, dual-remote CI/CD; enforced tenant isolation via router dependency + integration test suite.
- Designed a 12-agent test case generation pipeline (3 generator agents + validator-in-loop per team member) producing 3M+ labelled evaluation rows with strict held-out integrity guarantees, used as ground truth for scale-invariance validation across three orders of magnitude.

Notebrain Software & Services Pvt. Ltd

Aug 2025 – October 2025

AI/ML Intern

- Built and evaluated ML models for sports outcome prediction integrated into a web application, owning the end-to-end pipeline from data preprocessing through feature engineering to model training and validation.
- Performed dataset cleaning and annotation across multiple data sources to prepare reliable training inputs, addressing missing values, inconsistencies, and class imbalance.
- Collaborated on iterative algorithm tuning and optimization in an agile workflow, contributing to measurable improvements in prediction performance.

Venture Launcher

June 2025 – July 2025

AI/ML Intern

- Performed data analysis and preprocessing for an investor matching system, including data cleaning, feature identification, and exploratory analysis.
- Built a prototype ML model for matching relevance as a proof of concept for the founding team.

Projects

SurfVigil — Real-Time Browser Threat Detection Extension | [Extension link](#)

- Shipped Chrome extension + FastAPI backend with a 4-model ML ensemble (phishing, XSS, cryptomining, DOM anomaly) + Google Safe Browsing pre-check. Deployed on Hugging Face Spaces + Neon Postgres; published on Chrome Web Store.
- Diagnosed and patched three silent ML model failures (sklearn version drift, dict-key mismatch, wrong API kwarg) that had been returning HTTP 200 with three of four models silently reporting zero for days before detection.
- Recalibrated ensemble decision thresholds against a 17-site live-traffic sample when the false-positive WARN rate hit ~65% on modern web pages; cut it to ~0% without losing any genuine detections.

Smart City Guide — Java Swing Desktop Application | [Releases V1.0](#)

- Built a 4-layer Java desktop app (Presentation, Navigation, Services, Data Access) backed by a 7-table MySQL schema via JDBC, covering city profiles, tourism, transport, and user accounts across 20 cities.
- Implemented secure authentication using PBKDF2-HMAC-SHA256 (120K iterations, per-record salt, constant-time comparison); eliminated SQL injection surface via Prepared Statement on every query.
- Designed self-bootstrapping first-run flow (auto-creates schema, seeds data, persists credentials) and packaged v1.0 as a Windows installer with bundled JRE via jpackage — zero manual setup for end users.

Rice Leaf Disease Detection System | [Website](#)

- Built a two-stage ensemble of 5 models — Stage 1 (EfficientNet-B3, DenseNet-121, MobileNetV3) for 7-class triage at 96.68% accuracy, Stage 2 (ViT-Base, ConvNeXt-Tiny) for bacterial disease refinement at 98.18% accuracy.
- Optimized pipeline from 6 models to 5, reducing inference time by 16% while improving accuracy by 0.91%.
- Deployed end-to-end via PyTorch + Flask on Hugging Face Spaces with REST API, Docker support, and sub-2s processing time.x

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, HTML, SQL, CSS

ML/DL: PyTorch, scikit-learn, XGBoost, LightGBM, SHAP, Pandas, NumPy

Backend & Databases: FastAPI, Flask, SQLAlchemy, REST APIs, JWT / bcrypt auth, PostgreSQL, MongoDB

Tools & Platforms: Git/GitHub, Docker, Hugging Face Spaces, GitHub Actions (CI/CD), Chrome Extension APIs (Manifest V3)

Core Concepts: Data Structures, Algorithms, OOP, Version Control, Model Deployment, Feature Engineering, Model Calibration, Multi-tenancy

Position of Responsibility

Core Member – UPES ACM & ACM-W Student Chapters

- Served as Program Head for *ICPC Code Nexus*, managing logistics and tech ops.
- Awarded Certificate of Appreciation for excellent execution and contribution.

Social Outreach Intern — Shri Brahma Ji Siksha Avam Samaj Kalyan Samiti (May 2024 – July 2024)

- Coordinated education and environment awareness drives across 30+ communities, conducting 150+ field surveys.